



Technical Service Bulletin: TSB240153

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New Turbocharger Compressor Wheel for Single Stage HE800 Mining Engines:
Turbocharger Compressor Wheel Splitting

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK50 CM2150 MCRS

Issue

Aluminium HE800 Turbocharger compressor wheels used in single stage mining applications have malfunctioned because of fatigue. The compressor wheel splits through the wheel hub centre line and damages the compressor cover, turbine wheel and shaft. Malfunctions can spread debris into the aftercoolers on the affected bank.

Symptom:

- Reports of low power and black smoke from the affected bank
- Turbocharger compressor wheel is found split through the hub center line

Root Cause:

- The existing turbocharger compressor wheel material can experience crack initiation and fatigue propagation under certain duty cycle loads.

Verification

- Verify low power, black smoke from the affected bank, and low intake manifold pressure.
- Turbocharger compressor inspection is required to verify the failure mode. The turbocharger compressor wheel will be split in two or three pieces (Figure 1).

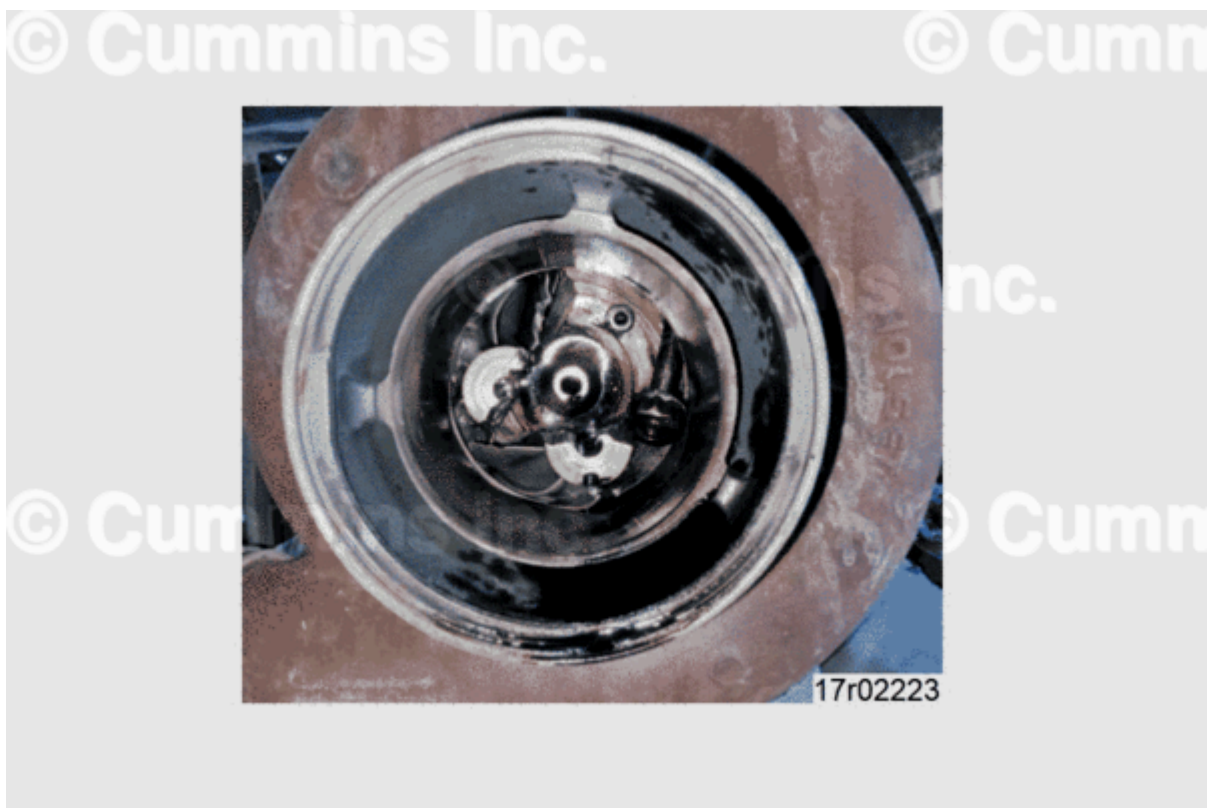


Figure 1, Turbocharger Compressor Wheel Split Malfunction.

Resolution

- Turbochargers **must** be replaced with new parts when the split compressor wheel failure mode has been verified. Remove aftercooler covers and clean or replace aftercoolers which are damaged or contaminated with compressor wheel debris.
- In the event of a turbocharger malfunction, the use of the new turbocharger part number is recommended. See Table 1.
- A new compressor wheel material has been used to improve HE800 robustness in single stage turbocharged mine haul truck and excavator applications. New turbocharger part numbers have been released for production and service. New turbochargers retain every feature of the previous part numbers, aside from the improved compressor wheel material.
- The new compressor wheel material has improved capability to resist increased stress at the compressor wheel to oil slinger interface. Increased stresses have been attributed to turbocharger resonance under certain running conditions.

Service Instructions

- If a turbocharger malfunction has been verified, remove aftercooler covers to inspect for debris. Clean the aftercooler housing and core if required. Install the aftercooler and covers. See QSK50 CM2150 MCRS Service Manual, Bulletin 4021528. Refer to Procedure 010-008 (/qs3/pubsys2/xml/en/procedures/28/28-010-008-tr.html) in Section 10.
- Replace turbocharger. See QSK50 CM2150 MCRS Service Manual, Bulletin 4021528. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html) in Section 19.

Service Parts Availability

- Service parts are available. See Table 1 for part numbers.

- Turbocharger availability for service will be in the form turbocharger kits **only**. The kit will include the new turbocharger, as well as the required gaskets and seals to replace one turbocharger.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
KIT, TURBOCHARGE R (Left Bank)	4956149	No	No	6524218
KIT, TURBOCHARGE R (Right Bank)	4956148	No	No	6524187

Part Identification

Turbocharger dataplates mounted to the compressor cover display part number details. For production units, **not** previously replaced, customer numbers (shown as “Cust” on the dataplate) can be used to identify turbocharger kit part numbers. For previously replaced turbochargers, a turbocharger kit part number will be shown on the turbocharger dataplate. See Table 2 for details.

Table 2, Customer Number to Kit Part Number Guide		
	Dataplate Customer Number	Related Turbocharger Kit
Existing Parts	4041284	4956149 (Left Bank)
	4041286	4956148 (Right Bank)
New Parts	6524217	6524218 (Left Bank)
	6524129	6524187 (Right Bank)

Production Status

Both new part numbers are implemented for production and were built into the ESN First. See Table 3.

Table 3, Production Information		
ESN First	Build Date ¹	Plant
33236236	8 July 2024	Daventry Engine Plant
¹ Engine build date can be found on engine dataplate.		

Document History

Date	Details
2024-7-24	Module Created
2025-1-7	Removed OEM and Products section. Updated Root Cause section.
2025-9-10	Updated Table 3 with proper ESN and build date information.

Last Modified: 10-Sep-2025
